

Sunghan Park

Engineering at Cambridge. Building tools for **crypto markets**, applied cryptography and ML.

Looking for internships in blockchain, ML and finance.

hugopark236@gmail.com

sunghanpark.vercel.app

github.com/hpark236

linkedin.com/in/sunghanpark

Singapore

EXPERIENCE

- Research Intern, UAV Research · A*STAR Institute for Infocomm Research

Apr 2024 - Aug 2024
- Ran Ubuntu and ROS 2 simulations to test and iterate drone perching mechanisms for forestry and aerospace tasks; co-developed a hybrid mounting leg and claw that improves quadcopter stability during aerial tasks.
- Financial Analysis Engineer · Wharton Global High School Investment Competition

Semifinalist, May 2024
- Built the team's financial analysis for a simulated client portfolio, covering company valuation, risk and allocation; the team reached the semifinal round.
- Head of Curriculum and Mentor · Code For All

Sep 2022 - Jan 2025
- Designed the Java, Python and Scratch curriculum for a nonprofit teaching underprivileged children in Singapore to code; mentored students and volunteer tutors and ran partnerships with PPIS, Care Corner and Impact.

EDUCATION

- University of Cambridge, Trinity College

From Oct 2027
- Engineering, Information Engineering track
- Singapore American School

Aug 2022 - May 2025
- High School Diploma. Mu Alpha Theta, Computer Science Honor Society, Science National Honor Society. Robotics, Research Club.

WEB3 PROJECTS

- xvenue-arb · tennis and golf arbitrage scanner

xvenue-arb.vercel.app
- Compares tennis and golf prices across Polymarket, Kalshi and Pinnacle including fees, flags when backing both outcomes on two venues costs under \$1, and sizes the trade from live order books.
- token-risk · memecoin health evaluator

token-risk.vercel.app
- Grades Solana and EVM tokens 0–100 on mint/freeze authority, honeypot and tax checks, liquidity and LP locks, holder concentration (Gini, HHI, insider clusters) and trade flow, with hard caps for disqualifying risks.
- secp256k1-js · secp256k1 cryptography from scratch

secp256k1-js.vercel.app
- Curve arithmetic, Keccak-256, RFC 6979 ECDSA with ecrecover, BIP-340 Schnorr and Shamir sharing in JavaScript BigInt, verified against OpenSSL and published vectors. Live demo extracts a private key from a reused nonce.
- cpamm-mev · Solidity constant-product pool and MEV simulator

cpamm-mev.vercel.app
- Uniswap-V2-style pool in Solidity (OpenZeppelin, Hardhat). Simulator reads mainnet reserves over raw JSON-RPC and solves optimal sandwich and CEX-arbitrage trades; contract tests replay the sandwich and match the model to the wei.

AI AND QUANT PROJECTS

- FinCouncil**, open-source multi-agent LLM investment council: five analyst agents plus a CIO synthesis on live brokerage data, with cross-examination and fact-checking.
- QUANTDESK**, paper-trading quant terminal (FastAPI): pluggable Python strategies, backtester, Binance and pump.fun memecoin feeds, alert scanner.
- Hold'em Noir**, poker engine whose bots track Bayesian ranges over 1,326 combos, with Monte Carlo equity and a push/fold Nash solver by fictitious play.
- SwingLab**, in-browser computer vision: MediaPipe pose and object detection, Hough-transform court detection, kinetic-chain analysis.

SKILLS

CODE	Python, JavaScript / TypeScript, Solidity, Java, SQL
WEB3	EVM and JSON-RPC, Hardhat, OpenZeppelin, AMMs and MEV, prediction-market microstructure, Solana token programs, ECDSA / Schnorr
ML / QUANT	MediaPipe, NumPy, pandas, Monte Carlo, Bayesian inference, backtesting, LLM agents, FastAPI, ROS 2
LANGUAGES	English and Korean (native), Mandarin and French (professional)